

Question number	Answer	Additional guidance	Mark
5(a)	An answer that includes two of the following points: - nuclear DNA is linear and mtDNA is circular (1) - nuclear DNA has unbound {phosphate group / sugar} whereas mtDNA does not (1) - mtDNA has (far)fewer {phosphodiester bonds / base pairs} (1)	DO NOT PIECE TOGETHER IGNORE straight / strands double helix IGNORE shorter / smaller ACCEPT other <u>structural</u> difference e.g. nuclear DNA organised around histones whereas mtDNA {is not / organised into nucleoids} nuclear DNA has 3' and 5' ends but mtDNA does not	(2)

Question number	Answer	Additional guidance	Mark
5(b)(i)	Any two from • (DNA) primers (1) • (DNA) (mono)nucleotides (1) • {taq / DNA} polymerase (1) • buffer (1)	NB Three correct = 2 marks, one or two correct = 1 mark ACCEPT all 4 listed IGNORE bases DO NOT ACCEPT RNA nucleotides ACCEPT taq enzyme DO NOT ACCEPT RNA polymerase	(2)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	• length of time calculated for the 35 cycles (1) • 1.38 (hours) (1)	4950 (seconds) / 82.5 (minutes) / 1.375 (hours) ecf if a given value in {seconds / minutes} is correctly converted into hours with 2 dps Bald answer of 1.38 = 2 marks Bald answer of 4950 (seconds) / 82.5 (minutes) / 1.375 (hours) = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
5(b)(iii)	An explanation that includes the following points: • {(high) temperature / 94°C} to {break the H bonds (between the strands) / separate the strands} (1) • {temperature lowered / 54°C} so that primers can attach (1) • {temperature raised (slightly) / 72°C} new (mono)nucleotides can {align / join} to bases (to form two molecules) (1) • {35 / many} cycles so that {several molecules / enough DNA} is made (1)	ACCEPT temperature lowered if no mention of primers ACCEPT appropriate temperature for (DNA) polymerase IGNORE bases / strands IGNORE strands NB Max 2 marks if no ref to change or quoted temperature from our diagram, provided the steps are in the correct order	(3)

Question number	Answer	Additional guidance	Mark
5(c)	An explanation that includes the following points: • by DNA profiling (1) • so that banding patterns (in each crow can be produce) and compared (1) • the more similar the profiles the more closely-related the crows (1) OR • by DNA profiling (1) • so that base sequences (in each crow can be determined) and compared (1) • the more similar the (base) sequences the more closely-related the crows (1)	ACCEPT (gel) electrophoresis / description of the process DNA fingerprint IGNORE common ancestor ACCEPT bioinformatics	(3)